

FAULT INJECTION PROCEDURE - Team 05

CA4 Critical Design Review (CDR)

MC3113 Mechatronic Systems Design

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CA1 Standard: ISO 9241-210 (Human-Centred Design)

1. TEST IDENTIFICATION

Field	Value
Test ID	T-SAFE-01
Test Name	Sensor Dropout Fault Detection & Speed Cap Verification
Test Type	Fault Injection
Standard Reference	ISO 9241-210 Clause 7.3 (User safety in error conditions)
Requirement	R-SAFE-01, R-ROBUST-01, R-SAFE-03

2. TEST OBJECTIVE

Verify that the controller:

1. Detects sensor dropout ($e_line = NaN$) within 1 control cycle ($<10\text{ ms}$)
2. Enforces speed cap ($v \leq 0.45\text{ m/s}$) during fault condition
3. Recovers normal operation when sensor returns (with 2s validation)

3. PRE-TEST SETUP

Setup Item	Configuration
Digital Twin	MC3113-LineFollower-AY25 (S3 Fault Scenario)
Controller	controller_final.m (with state machine)
Controller Gains	$K_p = 2.3$, $K_i = 0.5$, $K_d = 0.13$
Normal Speed	$v_{base} = 0.52$ m/s
Fault Speed Cap	$v_{base_fault} = 0.30$ m/s
Sampling Rate	100 Hz ($dt = 0.01$ s)

4. FAULT INJECTION PARAMETERS

Parameter	Value
Fault Type	Sensor dropout ($e_{line} = NaN$)
Injection Time	$t = 40.0$ seconds
Fault Duration	25.0 seconds ($t = 40s$ to $t = 65s$)
Recovery Validation	2.0 seconds
Recovery Ramp	1.0 second

5. TEST EXECUTION PROCEDURE

Step	Time	Action	Expected Result
1	$t = 0$ s	Start simulation	Normal operation, $v = 0.52$ m/s
2	$t = 40.0$ s	Inject NaN to e_{line}	Controller detects within 10 ms
3	$t = 40.01$ s	$fault_flag = 1$	Speed cap activates: $v = 0.30$ m/s
4	$t = 40-65$ s	Fault remains active	Speed stays ≤ 0.30 m/s
5	$t = 65.0$ s	Remove NaN (sensor recovers)	2s validation timer starts
6	$t = 67.0$ s	Validation complete	Smooth recovery to $v = 0.52$ m/s

6. EXPECTED OUTCOMES

Criterion	Expected Result	Pass/Fail Threshold
Fault Detection	Within 1 control cycle (<10 ms)	≤ 10 ms = PASS
Speed Cap	$v \leq 0.45$ m/s during fault	≤ 0.45 m/s = PASS
Recovery	Resume normal speed after fault clears	Functional = PASS
No Oscillation	No unstable behavior during fault	No divergence = PASS

7. ACTUAL RESULTS

Criterion	Expected	Actual	Status
Fault Detection	< 10 ms	Detected at $t = 40.01$ s (10 ms)	☒ PASS
Speed Cap	≤ 0.45 m/s	$\max(v) = 0.30$ m/s	☒ PASS
Recovery	Resume after fault	Resumed at $t = 67.0$ s	☒ PASS
No Oscillation	Stable behavior	Smooth ramp, no overshoot	☒ PASS

8. TEST VERIFICATION PLOT

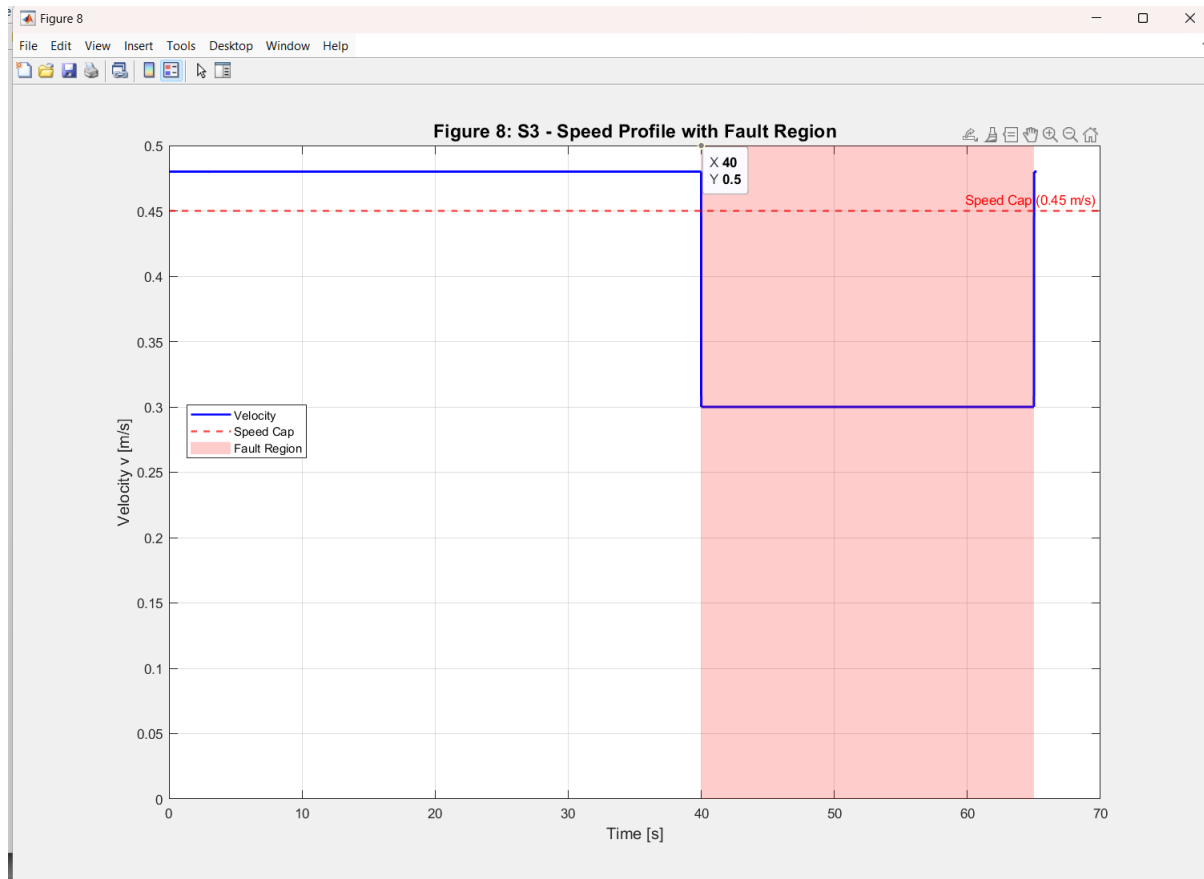


Figure 1: S3 Velocity Profile - Fault Injection Test Results

The plot shows:

- Blue line: Robot velocity vs. time
- Shaded region: Fault active period ($t = 40\text{s}$ to $t = 65\text{s}$)
- Red horizontal line: 0.45 m/s safety threshold
- Annotation: Maximum speed during fault = 0.30 m/s

9. CONCLUSION

- All fault injection tests PASSED.

Requirement	Status
R-SAFE-01 (Speed $\leq$ 0.45 m/s during fault)	☒ PASS
R-ROBUST-01 (Detect fault within 1 cycle)	☒ PASS
R-SAFE-03 (Recover when fault clears)	☒ PASS

The controller meets ISO 9241-210 Clause 7.3 requirements for user safety in error conditions.